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Research report

Age-related changes in cerebral blood flow and glucose metabolism in conscious rhesus monkeys

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Abstract

Regional cerebral blood flow (rCBF) and regional cerebral metabolic rate of glucose (rCMRglc) were measured in aged and young monkeys by positron emission tomography (PET). Our purpose was to examine whether the age-related changes observed in the human brain also occur in the monkey brain. Studies were performed on six aged and six young-adult male rhesus monkeys (*Macaca mulatta*). rCBF and the rCMRglc were serially measured using PET with [^{15}O]H_2O and 2-[^{18}F]fluoro-2-deoxy-D-glucose (FDG), respectively. In order to minimize the bias induced by anesthesia, the PET emission scans were performed in the conscious state. ROIs were taken for the cerebellum, hippocampus with adjacent cortex, striatum, occipital cortex, temporal cortex, frontal cortex and cingulate. Group differences and correlations between rCBF and rCMRglc in each group were determined. Aged monkeys had significantly lower rCBF in the cerebellum, hippocampus with the adjacent cortex, striatum, occipital cortex, temporal cortex, frontal cortex, and significantly lower rCMRglc in the cerebellum, hippocampus with the adjacent cortex, striatum, occipital cortex, temporal cortex, frontal cortex and cingulate, compared to young monkeys. There were significant correlations between rCBF and rCMRglc in both the aged and young groups, but no significant difference was found in relationship between the two groups. Age-related changes were observed not only in rCMRglc, but also in rCBF in aged monkeys, while the coupling between rCBF and rCMRglc was maintained even in aged monkeys. These results demonstrated the potential of aged monkeys to serve as an aged human model using PET. © 2002 Elsevier Science B.V. All rights reserved.

Theme: Neural basis of behavior

Topic: Aging

Keywords: Positron emission tomography; Cerebral blood flow; Cerebral metabolic rate of glucose; Rhesus monkey; Aging

1. Introduction

Numerous studies have been reported using positron emission tomography (PET) with normal aged human and Alzheimer's disease (AD) patients. In these PET studies, regional cerebral blood flow (rCBF) and regional cerebral metabolic rate of glucose (rCMRglc) were regarded as important indices of brain function, because they were revealed to correlate with pathological changes [4,9] and also reflected cognitive function [12,14,25]. Studies in normal aged human subjects showed age-related changes in brain glucose metabolism that decreased with aging [2,3,15,17] and recent studies showed regional loss in the frontal and temporal lobes [3,17]. The pattern of glucose metabolism in normal aged humans is almost the same as that of early AD patients, but a marked difference in the posterior cingulate cortex, site of the highest human rCMRglc activity, was found [23,24]. Studies in age-related changes in human rCBF were also reported. Martin

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et al. reported age-related decreases in rCBF in association with limbic cortices, which may subserve higher-order function such as cognition, memory, attention and control of behavior [19]. Leenders et al. reported rCBF decreased with age approximately 0.50% per year in cortical regions, while the decline in other regions was less evident [16]. Pantano et al. also found gray matter rCBF decreased linearly with age [28]. These studies have proved rCBF and rCMRglc decrease in human aging.

In recent years, PET studies in aged animal models have been performed with advances in high-resolution PET scanners. While there is no completely satisfactory model of human aging, the aged monkey could be a good model because of its behavioral and pathological characteristic features [30]. Eberling et al. reported that aged monkeys had a generally lower rCMRglc than younger monkeys, with the largest difference in the left temporal cortex [6]. Although rCMRglc is closely coupled to local neuronal activity and rCBF is continually adjusted to meet the dynamic alterations in local metabolic demand associated with normal physiological events [33,41], the fundamental relationship is affected by altered pathophysiological states [33] or pharmacological manipulation [1]. Therefore the relationship might be altered in age-related changes in the monkey brain. Tsukada et al. reported an impairment of the coupling mechanism between neuronal activation and rCBF response in the aged rhesus monkey [39], and its improvement with docosahexaenoic acid (DHA) [37]. This suggests that the relationship among rCBF, rCMRglc and neuronal activity appears to change with aging. However, to date, there have been only a few PET studies in aged monkeys as described above, due to the rarity of aged monkeys and also to the technical difficulties associated with PET scanning in conscious monkeys. There should be more investigations to elucidate age-related changes occurring in monkey brains.

In the present study, a sophisticated PET measurement procedure [27,35] was employed that enabled us to measure rCBF with $[^{15}O]H_2O$, a non-metabolic-trapping-type tracer in aged conscious monkeys. It also required habituation to a chair restraint to exclude the effect of stress in this study. We confirmed their habituation effect in training by measuring cortisol levels in plasma as a marker of stress (data not shown). Therefore, in the present study, the effect of anesthesia, as well as restraint stress, was minimized and age-related changes in conscious monkeys were examined in both rCBF and rCMRglc using PET, as well as the relationship between rCBF and rCMRglc. Our present study was the first to investigate age-related changes in rCBF in conscious monkeys and the results provide beneficial information for animal PET studies in aged monkey as a useful model of human aging. In the future, the reduced rCBF and/or rCMRglc are expected to be indices of medication efficacy in a drug evaluation study, utilizing PET with aged monkey as an aged human model.

2. Materials and methods

2.1. Animals

Studies were performed on six aged (mean age, 20.9±1.2 years; mean body weight, 7.4±0.7 kg) and six young-adult (mean age, 6.2±0.8 years; mean body weight, 6.0±0.8 kg) male rhesus monkeys (*Macaca mulatta*). Monkeys were maintained and handled in accordance with the recommendations of the United States National Institute of Health. T1-weighted MR-images of monkeys were obtained on an MRT-50A/II scanner (Toshiba, Tokyo, Japan) with a magnetic field strength of 0.5 T. Each monkey was chronically attached to an acrylic plate head-holder by the skull [27,35], which was used to fix each monkey's head to a monkey chair during PET scans. The animal was allowed to recover from the procedure for more than 1 month. Then, the monkey was habituated to a chair restraint several times a week for more than 1 month before the initiation of PET.

2.2. PET experiment

PET scans were performed with a high-resolution animal PET scanner (SHR-7700; Hamamatsu Photonics K.K., Hamamatsu, Japan), with transaxial resolution of 2.6 mm full-width at half-maximum (FWHM) in the center of the scan field, and a center-to-center distance of 3.6 mm [40]. A ^{68}Ga–^{68}Ge-blank scan (120 min) was performed before each study. During catheterization of the left femoral artery to measure mean arterial blood pressure (MABP), heart rate and arterial blood sampling, and catheterization of the saphenous vein for administration of tracers, monkeys were transiently anesthetized with about 2% sevoflurane in a N_2O/O_2 gas mixture (N_2O:O_2=7:3). After catheterization, anesthesia was immediately discontinued. Each monkey's head was fixed to a monkey chair for PET scans with a head-holder and stereotaxically aligned parallel to the orbito-meatal (OM) plane with a laser marker. Then, a ^{68}Ga-^{68}Ge-transmission scan (30 min) was performed. From the start of each transmission scan, MABP and heart rate were continuously monitored with a life monitoring system (Nihon Kohden, Tokyo, Japan). More than 1 h after discontinuing anesthesia, the $PaCO_2$, PaO_2 and pH of arterial blood were measured (STAT PROFILE blood gas analyzer: NOVA Biomedical, Waltham, MA) to confirm the physiological condition of the monkey. PET emission scans were performed with the monkey's ears unplugged, eyes open and in a dimmed light.

For the rCBF measurement, $[^{15}O]H_2O$ (786±45 MBq in 2 ml of saline) was injected intravenously. A 2-min emission scan consisted of 12 frames, was obtained following injection of $[^{15}O]H_2O$. During that period, 24 timed arterial blood samples (0.2 ml) were withdrawn

from a catheter placed in a femoral artery to measure arterial radioactivity using an ARC-2000 auto well gamma counter (Aloka, Tokyo, Japan).

Because of the very short half-life of ^{15}O (2.037 min), the radioactivity rapidly decayed about 15 min after the $[^{15}O]H_2O$ study, and FDG (419±18 MBq in saline of 2 ml) was injected intravenously for the rCMRglc measurement. A 60-min emission scan consisting of 22 frames, was obtained following injection of FDG. During that period, 16 timed arterial blood samples (0.2 ml) were withdrawn for arterial radioactivity measurements. The blood samples obtained at 45 and 60 min after injection were also analyzed for blood glucose levels with COBAS READY (Nihon Roche, Tokyo, Japan).

2.3. Data analysis

Images of rCBF were generated in accordance with an autoradiographic method [10,13,31]. PET images from 0 to 90 s after injection of $[^{15}O]H_2O$ and an arterial input function were used to calculate rCBF images. A partition coefficient of 0.7, derived from preliminary kinetic analysis in young monkeys (data not shown), was used for both young and aged monkeys.

Images of rCMRglc were generated in accordance with an autoradiographic method with an operational equation derived by Sokoloff et al. [34] and modified by Phelps et al. [29]. The parameters derived by Reivich et al. were used for this calculation [32]. PET images from 40 to 60 min after injection of FDG were used to calculate rCMRglc images.

Regions of interest (ROIs) were manually drawn on MRI images for each monkey. Stereotaxic coordinations of PET and MRI were adjusted based on the OM plane with a head-holder attached to the head of each monkey. ROIs were taken for the cerebellum, hippocampus with adjacent cortex, temporal cortex, occipital cortex, striatum, frontal cortex and cingulate. The rCBF and rCMRglc were obtained for these ROIs. Each ROI of the cerebrum was also divided into two parts, that is, the right and left for the hippocampus with the adjacent cortex, temporal cortex, occipital cortex, striatum and frontal cortex. The ROI for the cingulate was divided into anterior and posterior, because the human posterior cingulate has the highest activity which decreases even in very early AD [23,24].

All data are represented as means±S.D. Analysis of variance (ANOVA) with repeated measures was used to test for statistical differences between the aged and young groups. ANOVA followed by Scheffe's post-hoc test were used to test for statistical differences between groups at individual ROIs. Correlation analysis was performed between rCBF and rCMRglc in the aged and young groups according to the method described by McCulloch et al. [22]. Differences were considered significant when $P<0.05$.

Table 1
Physiological parameters of the monkeys

	Young monkeys	Aged monkeys
HR (bpm)	192±46	196±39
MABP (mmHg)	113±17	108±18
pH	7.460±0.021	7.460±0.016
pCO_2 (mmHg)	35.1±19	34.4±1.9
pO_2 (mmHg)	92.4±2.6	94.5±5.2
Blood Glucose (mg/dl)	102±37	103±27

Values are mean±S.D. HR, heart rate; MABP, mean arterial blood pressure.

3. Results

Monkeys subjected to PET scanning procedures showed no overt behavioural signs of stress. Similarly, cardiovascular and respiratory variables measured immediately prior to the scans were within normal ranges in both the aged and young groups (Table 1).

ANOVA with repeated measures revealed group differences in rCBF between aged and young groups for both data of averaged ROIs ($F(1,10)=15.1$, $P<0.005$) and data of divided ROIs ($F(1,10)=13.4$, $P<0.005$). Post-hoc analysis for these data revealed ROIs that reached statistical significance as shown in Table 2. ANOVA with repeated measures also revealed group differences in rCMRglc between aged and young groups for both data of averaged ROIs ($F(1,10)=11.6$, $P<0.01$) and data of divided ROIs ($F(1,10)=11.1$, $P<0.01$). Post-hoc analysis for these data revealed ROIs that reached statistical significance as shown in Table 3. Differences between hemispheres for each ROI did not reach statistical significance.

The rCBF in each ROI was significantly correlated with the corresponding rCMRglc in both aged ($r^2=0.6195$, $P<0.0001$) and young ($r^2=0.6253$, $P<0.0001$) monkeys. Fig. 1 shows the correlation between the rCBF and the rCMRglc from all data of averaged ROIs. Regression lines for each group are also shown. The difference between these two relationships was not statistically significant ($P=0.31$).

4. Discussion

The main findings of the present study were that rCBF was reduced in aged monkeys compared with young monkeys and no marked regional difference in the degree of reduction, as observed in human frontal and temporal cortex, was found. In humans, the highest rCMRglc activity was in posterior cingulate [23,24]. However, in the present study, there was no such marked high activity in the cingulate, while relatively high activity appeared in the occipital cortex in young monkeys but not in aged monkeys. A fundamental relationship between blood flow and underlying metabolic activity was maintained. The present

Table 2
Regional cerebral blood flow in young and aged monkeys

	Young monkeys (ml/100 g/min, mean±S.D.) (*Y*)			Aged monkeys (ml/100 g/min, mean±S.D.) (*A*)			*Y–A* (%)		
	Average	R or A	L or P	Average	R or A	L or P	Average	R or A	L or P
Cerebellum	47.8±5.1			34.3±7.6			28[b]		
Hippocampus with adj. cortex	53.9±7.6	53.2±7.2	54.6±8.3	43.4±4.6	43.3±4.0	43.5±5.5	20[a]	19[a]	20[a]
Striatum	60.5±11.1	60.2±12.7	60.9±10.4	44.1±3.9	43.8±4.0	44.5±4.1	27[b]	27[a]	27[b]
Occipital cortex	59.2±7.6	59.6±8.2	58.9±7.1	43.9±7.2	44.0±6.9	43.8±7.6	26[c]	26[b]	26[b]
Temporal cortex	53.2±6.5	54.7±3.0	51.6±2.5	45.7±4.6	43.9±1.7	47.6±2.1	14[a]	20[a]	8
Frontal cortex	45.0±6.4	45.2±3.1	44.8±2.5	37.2±3.6	35.7±1.5	38.7±2.8	17[a]	21[a]	14
Cingulate	54.5±9.6	54.5±4.4	54.5±3.5	46.5±2.1	46.3±0.7	46.8±1.3	15	15	14

$Y–A = (Y–A)/Y \times 100$. Average, values are averaged for the right and left hemisphere, except the cingulate. For the cingulate, values are averaged for the anterior and posterior cingulate; R or A, right hemisphere except for the cingulate; anterior cingulate. L or P, left hemisphere except for the cingulate; posterior cingulate.
[a] $P<0.05$.
[b] $P<0.01$.
[c] $P<0.005$.

study is the first report to investigate age-related changes in both rCBF and rCMRglc, as well as in their coupling, in conscious monkeys. Though it is well known that rCMRglc reflects neuronal activity and rCBF is closely coupled to rCMRglc [33,41], Ohata et al. reported an alteration of the coupling which occurred in aged rats [26]. In the present study, the degree of rCMRglc reduction seemed to be a little greater than that of rCBF reduction (Tables 2 and 3). It also appears from Fig. 1 that the regression line for the aged group (dashed line) had a lower intercept. However, the difference between the two groups was not statistically significant, and our data did not reveal an alteration in the relationship between rCBF and rCMRglc with aging.

In animal PET studies, anesthetics are usually used, because it is difficult to keep the animal's head still during a PET measurement. However, the conscious condition is necessary to measure brain function, because anesthetic agent affects brain activity in an unpredictable fashion [11]. Inhalational anesthetic agents, such as isoflurane, sevoflurane and halothane were reported to have an intrinsic vasodilative effect that caused secondary abolishment of flow-metabolism coupling in humans [20,21] and rhesus monkeys [8]. Anesthesia with ketamine was also reported to significantly affect in vivo binding of [^{11}C]*N*-methylspiperone [27] and [^{11}C]raclopride [36] to dopamine D_2 receptors in the striatum of the monkey brain. It was also reported that binding potential (BP) of [^{11}C]β-CFT was markedly increased by isoflurane anesthesia [38]. Therefore, brain function measured under anesthesia conditions may not indicate the normal basal value as measured under the conscious condition. Our data are considered to almost exclude the influence of anesthesia and to be reflect the normal basal value as closely as possible.

The present results raise several issues: Firstly, it is not easy to classify or categorize an 'aged monkey' to a corresponding state in humans. Rhesus monkeys were reported to develop age-associated impairments in per-

Table 3
Regional cerebral glucose metabolism in young and aged monkeys

	Young monkeys (μmol/100 g/min, mean±S.D.) (*Y*)			Aged monkeys (μml/100 g/min, mean±S.D.) (*A*)			*Y–A* (%)		
	Average	R or A	L or P	Average	R or A	L or P	Average	R or A	L or P
Cerebellum	26.8±5.9			16.3±6.8			39[a]		
Hippocampus with adj. cortex	32.2±5.2	32.3±5.0	32.1±5.5	21.2±4.0	21.1±3.6	21.3±4.7	34[c]	35[c]	34[c]
Striatum	40.9±9.1	40.3±8.9	41.5±9.3	26.5±5.6	26.6±5.9	26.5±5.4	35[b]	34[a]	36[b]
Occipital cortex	39.7±9.4	40.1±10.0	39.3±8.9	26.7±7.7	27.0±8.0	26.4±7.5	33[a]	33[a]	33[a]
Temporal cortex	36.0±7.3	36.7±3.0	35.3±3.1	26.6±5.0	26.0±2.0	27.3±2.1	26[a]	29[a]	23
Frontal cortex	34.1±8.2	34.0±3.4	34.2±3.4	22.4±4.7	21.5±1.8	23.2±2.3	34[a]	37[b]	32[a]
Cingulate	36.6±8.5	36.2±3.6	37.1±3.4	26.7±4.5	26.3±1.4	27.1±2.4	27[a]	27[a]	27[a]

$Y–A = (Y–A)/Y \times 100$. Average, values are averaged for the right and left hemisphere, except the cingulate. For the cingulate, values are averaged for the anterior and posterior cingulate. R or A, right hemisphere except for the cingulate; anterior cingulate. L or P, left hemisphere except for the cingulate; posterior cingulate.
[a] $P<0.05$.
[b] $P<0.01$.
[c] $P<0.005$.

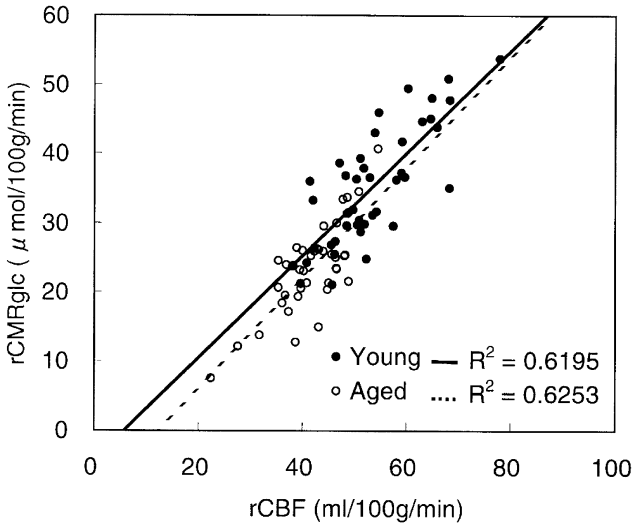


Fig. 1. Individual rCBF and rCMRglc for averaged ROIs of young and aged monkeys plotted with regression lines for these data. Correlation coefficients (R^2) are shown for each group.

formance on cognitive/memory tasks, and in their late teens, some animals show impairments in performance of certain spatial abilities, while senile plaques and amyloid deposits also appeared [30]. Eberling et al. also reported an age-dependent memory impairment and rCMRglc decline in aged rhesus monkeys (19–33 years of age) [7]. Subjects in the present study were 18–25 years of age; roughly equivalent to humans of perhaps 54–75 years of age. Some of them seemed to be a little young, but had at least developed some age-associated impairment. Subject selection with cognitive tasks might be useful to obtain more evident results.

Secondly, another issue was the possible effects of restraint stress on conscious measurements. Cardiovascular and respiratory parameters are an index of stress and our data shown in Table 1 show no overt signs of stress. This suggests 1 month of habituation successfully excluded the effect of stress in this study. Also, in a separate experiment, we tried to confirm their habituation effect in training by measuring cortisol levels in plasma as a marker of stress. The result showed 1 week training reduced plasma cortisol levels by about 75% of the initial level, and then the achieved level was maintained for a month by continuous training (data not shown). Therefore, we believe that the effect of restraint stress was minimized in the present study.

The last issue was an asymmetry in rCBF and rCMRglc reduction between the two hemispheres in aged monkeys, although some studies in normal aged [17] and dementia patients [5,18] demonstrated asymmetry in the degree of metabolic decline between hemispheres. Several possibilities, such as left-handedness, greater susceptibility of the left hemisphere to degenerative or ischemic brain disease, a sampling effect of neuropsychological testing, and variance among subjects may underlie the reported asymmetry. However, these issues are out of the scope of the present study. While Eberling et al. reported the largest difference in the left temporal cortex of aged rhesus monkeys [6], the present results did not show any significant difference between hemispheres. The asymmetry in the degree of reduction in rCBF and rCMRglc, as well as regional differences, in aged monkeys need to be addressed with a larger number of subjects in a future study.

5. Conclusion

In conclusion, we demonstrated rCBF as well as rCMRglc in aged monkeys were generally lower than young monkeys. We also demonstrated that a relationship between rCBF and rCMRglc was maintained, even in aged monkeys, and could not find a significant alteration in the relationship between aged monkeys compared with young monkeys. The results will facilitate animal PET studies in aged monkeys, which is a useful model of human aging, and may extend to the potential application of aged monkeys to drug evaluation studies using PET.

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